

2

Electromagnetic wave propagation

2.1 Maxwell's equations

Because reflectance spectroscopy uses electromagnetic radiation to probe matter, this book begins with Maxwell's electromagnetic equations and the solutions to them that describe propagating plane waves. For those whose knowledge of vectors may be a bit rusty, a brief review of vector notation is provided in Appendix A.1.

In their general form, Maxwell's equations can be written as follows:

$$\operatorname{div} \mathbf{D}_e = \rho_e, \quad (2.1)$$

$$\operatorname{div} \mathbf{B}_m = 0, \quad (2.2)$$

$$\operatorname{curl} \mathbf{E}_e = -\partial \mathbf{B}_m / \partial t, \quad (2.3)$$

$$\operatorname{curl} \mathbf{H}_m = \mathbf{j}_e + \partial \mathbf{D}_e / \partial t. \quad (2.4)$$

In these equations, $\mathbf{E}_e$ is the electric field, $\mathbf{D}_e$ is the electric displacement, $\mathbf{B}_m$ is the magnetic-induction field, $\mathbf{H}_m$ is the magnetic intensity, ρ_e is the electric-charge density, $\mathbf{j}_e$ is the electric current density, t is the time, and “div” and “curl” are, respectively, the vector divergence and curl operators. The reader is referred to the many excellent textbooks on electromagnetic theory for detailed derivations and more rigorous discussions of these equations, including Stratton (1941), Panofsky and Phillips (1962), Marion (1965), Elliott (1966), Landau and Lifschitz (1975), and Jackson (1999).

Equation (2.1) states that electric charges can generate electric fields and that the field lines diverge from or converge toward the charges. Equation (2.2) states that there are no sources of magnetic fields that are analogous to electric charges; that is, magnetic monopoles do not exist. According to equation (2.3), electric fields can also be generated by magnetic fields that change with time, and electric fields generated in this manner tend to coil or curl around the magnetic-field lines. Similarly, according to equation (2.4), magnetic fields can be generated by both electric

currents and time-varying electric fields, and the magnetic lines of force tend to curl around these sources.

Equations (2.1) – (2.4) are called the *field equations*. In order to solve them, additional relations, known as *constitutive equations*, that connect two or more of the quantities in the field equations are needed, along with appropriate boundary conditions. The constitutive relations describe the way that microscopic charges and currents generated in the medium by the applied field alter those fields. In the remainder of this chapter, solutions of the field equations that describe propagating plane waves will be discussed for several different types of constitutive equations.

2.2 Electromagnetic waves in free space

2.2.1 The wave equation

The simplest case of electromagnetic radiation is wave propagation in a vacuum. In free space there are no charges or currents, so the constitutive equations have the simple form

$$\rho_e = 0, \quad (2.5)$$

$$\mathbf{j}_e = 0, \quad (2.6)$$

$$\mathbf{D}_e = \varepsilon_{e0} \mathbf{E}_e, \quad (2.7)$$

$$\mathbf{B}_m = \mu_{m0} \mathbf{H}_m, \quad (2.8)$$

where ε_{e0} is the permittivity of free space ($\varepsilon_{e0} = 8.85 \times 10^{-12}$ coulomb²/newton-meter²), and μ_{m0} is the permeability of free space ($\mu_{m0} = 12.57 \times 10^{-7}$ amperes/m²). Then the field equations become

$$\text{div } \mathbf{E}_e = 0, \quad (2.9)$$

$$\text{div } \mathbf{B}_m = 0, \quad (2.10)$$

$$\text{curl } \mathbf{E}_e = -\partial \mathbf{B}_m / \partial t, \quad (2.11)$$

$$\text{curl } \mathbf{B}_m = \mu_{m0} \varepsilon_{e0} \partial \mathbf{E}_e / \partial t. \quad (2.12)$$

These equations may be combined to yield relations that each contain only one field quantity, as follows. Take the curl of both sides of (2.11) and use vector identity (A.9) from Appendix A:

$$\text{curl}(\text{curl } \mathbf{E}_e) = \text{grad}(\text{div } \mathbf{E}_e) - (\text{div} \cdot \text{grad}) \mathbf{E}_e = -\partial(\text{curl } \mathbf{B}_m) / \partial t. \quad (2.13)$$

Using (2.9) and (2.12) in (2.13) gives

$$\nabla^2 \mathbf{E}_e - \mu_{m0} \varepsilon_{e0} \partial^2 \mathbf{E}_e / \partial t^2 = 0, \quad (2.14)$$

where $\nabla^2 = \text{div} \cdot \text{grad}$ is the Laplacian operator (often called “del-squared”). Applying a similar analysis starting with the magnetic field, equation (2.12), yields

$$\nabla^2 \mathbf{B}_m - \mu_{m0} \epsilon_{e0} \partial^2 \mathbf{B}_m / \partial t^2 = 0. \quad (2.15)$$

Equations (2.13) and (2.14) are of the general form

$$\nabla^2 f = (1/v^2) \partial^2 f / \partial t^2. \quad (2.16)$$

Equation (2.16) is known as the wave equation and describes a disturbance at position $\mathbf{r}$ of arbitrary shape propagating with velocity v . If the geometry of the situation has plane-parallel symmetry, the general solution of (2.16) is $f = f(K\varnothing)$, where K is any constant, $\varnothing = \mathbf{u}_p \cdot \mathbf{r} \pm vt$ is the phase, $\mathbf{u}_p$ is a unit vector parallel to the direction of propagation, and f is an arbitrary function. All points satisfying $\mathbf{u}_p \cdot \mathbf{r} = \text{constant}$ lie on a plane perpendicular to $\mathbf{u}_p$ and passing through the point $\mathbf{r}$ (Figure 2.1).

In the remainder of this chapter, only plane waves will be considered. Without loss of generality, the direction of propagation may be taken to lie along the z -axis. In this case, f is independent of x and y , the differential vector operator “del” is $\nabla = \mathbf{u}_z \partial / \partial z$, where $\mathbf{u}_z$ is a unit vector pointing in the z direction, and $\varnothing = z \pm vt$. Then (2.16) becomes

$$\partial^2 f / \partial z^2 - (1/v^2) \partial^2 f / \partial t^2 = 0. \quad (2.17)$$

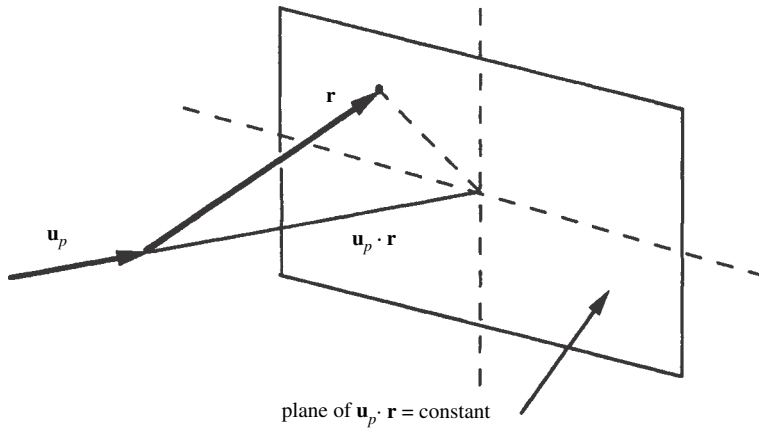


Figure 2.1

The proof that this equation is satisfied by any arbitrary function $f(K\varnothing)$ is almost trivial:

$$\frac{\partial^2 f}{\partial z^2} = \frac{d}{d\varnothing} \left(\frac{df}{d\varnothing} \frac{\partial \varnothing}{\partial z} \right) \frac{\partial \varnothing}{\partial z} = K^2 \frac{d^2 f}{d\varnothing^2}, \quad (2.18)$$

$$\frac{1}{v^2} \frac{\partial^2 f}{\partial t^2} = \frac{1}{v^2} \frac{d}{d\varnothing} \left(\frac{df}{d\varnothing} \frac{\partial \varnothing}{\partial t} \right) \frac{\partial \varnothing}{\partial t} = K^2 \frac{d^2 f}{d\varnothing^2}. \quad (2.19)$$

If $\varnothing = z - vt$, f represents a pattern whose shape is described by $f(K\varnothing)$ moving in the positive z direction with velocity v ; if $\varnothing = z + vt$, the pattern is moving toward negative z .

If the propagating pattern is periodic and repeats over a distance λ , then at any point z as the wave passes its pattern will repeat over a time interval $t = \lambda/v$. This interval is called the period. The reciprocal of the period is the frequency ν , and λ is the wavelength. It is convenient to let $K = 2\pi/\lambda$, so that $f(K\varnothing) = f(2\pi z/\lambda \pm 2\pi \nu t)$. Thus, v and ν are related by

$$v = \nu \lambda. \quad (2.20)$$

A large class of problems can be solved by representing the waves by sinusoidally varying functions of the form $f = f_0 \sin 2\pi(z/\lambda \pm \nu t)$ or $f = f_0 \cos 2\pi(z/\lambda \pm \nu t)$. The reason sinusoidal solutions are so useful is that the principles of Fourier analysis and synthesis allow an arbitrary function to be described mathematically by sums of sinusoidal waves of appropriate amplitude and phase, provided the function is mathematically well behaved. In practice this qualification is not greatly restrictive, because virtually any function that describes a physically real quantity will be well behaved.

Because exponential functions are easy to manipulate mathematically, it is often convenient to use complex variables to represent the sinusoidal functions through the relation

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (2.21)$$

where $i = \sqrt{-1}$ and θ is some arbitrary quantity. Once the complex solution has been obtained, only the real part represents an actual physical quantity. Thus, solutions of the wave equation of the form $\mathbf{E}_e = \mathbf{E}_{e0} e^{2\pi i(z/\lambda - \nu t)}$ and $\mathbf{B}_m = \mathbf{B}_{m0} e^{2\pi i(z/\lambda - \nu t)}$ may be investigated without loss of generality. Comparing (2.14) and (2.15) with (2.16) shows that these expressions for the fields represent plane sinusoidal waves moving in the positive z direction with velocity

$$v = (\mu_{m0} \epsilon_{e0})^{-1/2} = c_0, \quad (2.22)$$

where c_0 is the velocity of light in free space ($c_0 = 2.998 \times 10^8 \text{ m s}^{-1}$). The quantities $\mathbf{E}_{e0}$ and $\mathbf{B}_{m0}$ are the amplitudes of the fields and are determined by the boundary conditions.

The electric and magnetic fields are not independent, but are related through (2.11), which for a plane sinusoidal wave propagating in the z direction is

$$\text{curl } \mathbf{E}_e = \mathbf{u}_z \times \partial \mathbf{E}_e / \partial z = (2\pi i / \lambda) \mathbf{u}_z \times \mathbf{E}_e = -\partial \mathbf{B}_m / \partial t = 2\pi i v \mathbf{B}_m. \quad (2.23)$$

Thus,

$$\mathbf{B}_m = \mathbf{u}_z \times \mathbf{E}_e / c_0. \quad (2.24)$$

Similarly (2.12) is

$$\mathbf{u}_z \times \partial \mathbf{B}_m / \partial z = i (2\pi / \lambda) \mathbf{u}_z \times \mathbf{B}_m = \mu_{m0} \varepsilon_{e0} \partial \mathbf{E}_e / \partial t = -2\pi i v \mu_{m0} \varepsilon_{e0} \mathbf{E}_e, \quad (2.25)$$

so that

$$\mathbf{E}_e = -(1 / \lambda v \mu_{m0} \varepsilon_{e0}) \mathbf{u}_z \times \mathbf{B}_m = -c_0 \mathbf{u}_z \times \mathbf{B}_m. \quad (2.26)$$

Equations (2.24) and (2.26) show that $\mathbf{E}_e$ and $\mathbf{B}_m$ are perpendicular to each other and to the direction of propagation (Figure 2.2), and also that the amplitudes of the fields are related by

$$B_{m0} = E_{e0} / c_0. \quad (2.27)$$

Two independent orthogonal solutions to the wave equation are possible in which the electric vectors are perpendicular to each other. If the positive x direction is chosen to be parallel to $\mathbf{E}_e$, then the component of $\mathbf{B}_m$ corresponding to this solution points in the positive y direction. Figure 2.2 illustrates this solution. If the positive y direction is chosen to be parallel to the electric vector, then the accompanying magnetic vector points in the negative x direction. In free space, neither $\mathbf{E}_e$ nor $\mathbf{B}_m$ has a component parallel to z .

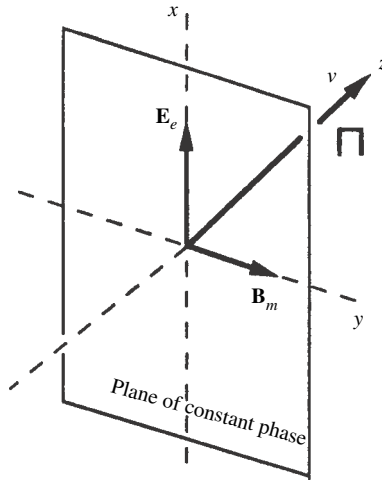


Figure 2.2 Relation between the fields and the propagation velocity vector in a plane electromagnetic wave.

2.2.2 Huygens's principle

Three hundred years ago when it was realized that light had a wave nature, the question arose as to what medium the waves were propagating through. It was postulated that all space was filled with an invisible fluid called the “aether.” However, in the modern view of an electromagnetic wave no aether is required. According to Maxwell's equations, a changing electric field generates a magnetic field, and a changing magnetic field generates an electric field. Thus, a propagating electromagnetic wave may be regarded as generating itself; that is, the changing fields on the wave front continually regenerate the wave. This concept leads to *Huygens's principle*, in which each point on a wave front may be considered to be a source of spherical wavelets that travel radially outward and combine coherently with wavelets from all the other points to produce a new wave front. If the wave front is plane and infinite in lateral extent, this process simply produces another plane wave front. However, if part of the wave front is obstructed, Huygens's principle predicts that fields still exist behind the obstructing object. This phenomenon is called *diffraction*, and Huygen's principle can be used to calculate the resultant fields in the vicinity of the object.

2.2.3 The Poynting vector and the irradiance

An important quantity, the power contained in the wave, can be obtained by forming the vector dot products of $\mathbf{E}$ with equation (2.12) and of $\mathbf{B}$ with (2.13),

$$\mathbf{E}_e \cdot \text{curl } \mathbf{B}_m = \mu_{m0} \varepsilon_e \mathbf{E}_e \cdot \partial \mathbf{E}_e / \partial t = (\mu_{m0} \varepsilon_{e0} / 2) \partial \mathbf{E}_e^2 / \partial t, \quad (2.28)$$

$$\mathbf{B}_m \cdot \text{curl } \mathbf{E}_e = -\mathbf{B} \cdot \partial \mathbf{B}_m / \partial t = (1/2) \partial \mathbf{B}_m^2 / \partial t; \quad (2.29)$$

subtracting gives

$$\mathbf{E}_e \cdot \text{curl } \mathbf{B}_m - \mathbf{B}_m \cdot \text{curl } \mathbf{E}_e = \frac{1}{2} \partial (\mu_{m0} \varepsilon_{e0} E_e^2 + B_m^2) / \partial t. \quad (2.30)$$

Using the vector identity, equation (A.10), this becomes

$$(1/\mu_{m0}) \text{div} (\mathbf{E}_e \times \mathbf{B}_m) = -\frac{1}{2} \partial (\varepsilon_{e0} E_e^2 + B_m^2 / \mu_{m0}) / \partial t. \quad (2.31)$$

Integrating both sides of the last equation over a volume V bounded by a closed surface A (Figure 2.3), and applying Gauss's theorem [equation (A.11)], gives

$$\begin{aligned} \int_V \text{div} (\mathbf{E}_e \times \mathbf{B}_m / \mu_{m0}) dV &= \int_A (\mathbf{E}_e \times \mathbf{B}_m / \mu_{m0}) \cdot d\mathbf{A} \\ &= -\frac{\partial}{\partial t} \int_V (\varepsilon_{e0} E_e^3 / 2 + B_m^2 / 2\mu_{m0}) dV. \end{aligned} \quad (2.32)$$

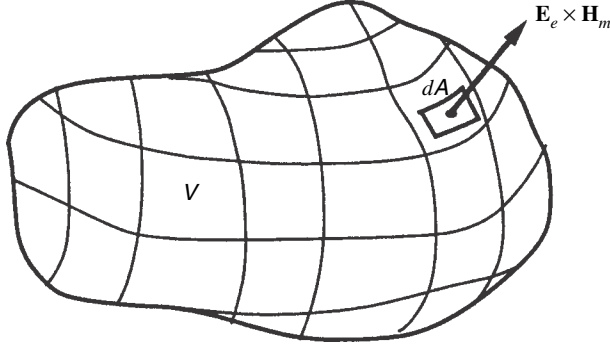


Figure 2.3

Now, $\epsilon_{e0} E_e^2/2$ and $B_m^2/2\mu_{m0}$ are, respectively, the energy densities in the electric and magnetic fields, so that the right-hand side of (2.32) describes the rate of decrease of energy density in V . Thus, the left-hand side must represent the outward movement of energy through the surface A bounding V . The quantity $\mathbf{E}_e \times \mathbf{B}_m/\mu_{m0}$ is the local flux of energy per unit area per unit time crossing surface element and is known as the *Poynting vector* Π ,

$$\Pi = \mathbf{E}_e \times \mathbf{B}_m/\mu_{m0} = \mathbf{E}_e \times \mathbf{H}_m = \mathbf{u}_z E_e^2/\mu_{m0}c_0. \quad (2.33)$$

The Poynting vector oscillates with time. The *irradiance* J of the wave is the average energy crossing unit area perpendicular to the direction of propagation per unit time. This quantity is often called the intensity, although the latter term is usually reserved for the power flux per unit solid angle. The irradiance is the time average (denoted by angular brackets) over one cycle of the product of the real parts of the fields in the Poynting vector,

$$\begin{aligned} \mathbf{u}_z J &= \langle \text{Re}(\mathbf{E}_e) \times \text{Re}(\mathbf{B}_m)/\mu_{m0} \rangle \\ &= \mathbf{u}_z (v/\mu_{m0}) \int_0^{1/v} E_{e0} B_{m0} \cos^2 2\pi(z\lambda - vt) dt. \end{aligned} \quad (2.34)$$

Using the relation $\cos^2 x = (1 + \cos 2x)/2$, this integral is readily evaluated to give

$$J = E_{e0} B_{m0}/2\mu_{m0} = E_{e0}^2/2\mu_{m0}c_0 = \sqrt{\epsilon_{e0}/\mu_{m0}} E_{e0}^2/2. \quad (2.35)$$

2.3 Propagation in a linear nonabsorbing medium

2.3.1 Induced electric dipoles

When fields are present in a medium, they act on the electrons and ions that make up the medium to induce electric and magnetic-dipole moments. The net electric charge ρ_e usually is zero because on the average the total number of positive charges

on the ions equals the total number of negative charges on the electrons. However, in response to the applied field the negative electron charge distribution may become slightly displaced from the positive ion charge distribution, thus generating local electric-dipole moments. The motions of these charges constitute currents, which generate magnetic-dipole moments. If the applied fields are time-varying, these induced electric and magnetic dipoles will in turn generate electromagnetic waves that will interact coherently with the incident wave and alter its propagation characteristics.

In many materials the local electric-dipole moment $\mathbf{p}_e$ induced in a medium by an applied electric field is linearly proportional to the field and is parallel to it. Thus, there is often a relation of the form

$$\mathbf{p}_e = \alpha_e \varepsilon_{e0} \mathbf{E}_{\text{loc}}, \quad (2.36)$$

where $\mathbf{E}_{\text{loc}}$ is the local electric field, and α_e is a constant of proportionality called the electric *polarizability*. The local field may be different from the general macroscopic field $\mathbf{E}_e$ in the medium.

The *electric polarization* $\mathbf{P}_e$ is the dipole moment per unit volume,

$$\mathbf{P}_e = N \mathbf{p}_e, \quad (2.37)$$

where N is the number of dipoles per unit volume. Often $\mathbf{P}_e$ is linearly proportional and parallel to the applied field $\mathbf{E}_e$, so that it can be described by a relation of the form

$$\mathbf{P}_e = \chi_e \varepsilon_{e0} \mathbf{E}_e, \quad (2.38)$$

where χ_e is the electric susceptibility. The electric displacement $\mathbf{D}_e$ is defined as

$$\mathbf{D}_e = \varepsilon_{e0} \mathbf{E}_e + \mathbf{P}_e = \varepsilon_{e0} (1 + \chi_e) \mathbf{E}_e. \quad (2.39)$$

The quantity in parentheses is the *dielectric constant* or specific inductive capacity,

$$K_e = 1 + \chi_e. \quad (2.40)$$

Thus,

$$\mathbf{P}_e = \varepsilon_{e0} (K_e - 1) \mathbf{E}_e, \quad (2.41)$$

and

$$\mathbf{D}_e = \varepsilon_e \mathbf{E}_e, \quad (2.42)$$

where

$$\varepsilon_e = \varepsilon_{e0} K_e \quad (2.43)$$

is the permittivity of the medium.

In general, ε_e may vary from place to place, as in an inhomogeneous or discontinuous medium. It may depend on the direction of $\mathbf{E}_e$, as in birefringent crystals.

In some materials $\mathbf{D}_e$ is not parallel to $\mathbf{E}_e$, in which case ε_e is a tensor and (2.42) is a matrix equation. The permittivity may be a function of $\mathbf{E}_e$, as often occurs when a substance is illuminated by an intense laser beam. However, in this chapter it will be assumed that ε_e is a homogeneous, isotropic, scalar constant. The problem of propagation in a discontinuous medium will be considered later in the book.

2.3.2 The Clausius–Mossotti / Lorentz–Lorenz relation

A question of considerable interest is the relation between the local field $\mathbf{E}_{\text{loc}}$ at the molecule, which defines α_e and $\mathbf{p}_e$, and the applied field $\mathbf{E}_e$, which determines K_e , ε_e , and $\mathbf{P}_e$. This is not a trivial question, and it has never been answered satisfactorily in general. It is discussed extensively in many advanced textbooks on electrostatics and electrodynamics.

Often it is simply assumed that $\mathbf{E}_{\text{loc}} = \mathbf{E}_e$. In that case, $\mathbf{P}_e = N\alpha_e\varepsilon_{e0}\mathbf{E}_{\text{loc}}$, $\chi_e = N\alpha_e$, and

$$K_e = 1 + N\alpha_e. \quad (2.44)$$

Another approach, which has considerable experimental support, is the following. Assume that at an arbitrary point the medium can be divided into two regions separated by a spherical surface of radius R and area $a = 4\pi R^2$ centered on some molecule having an induced dipole moment $\mathbf{p}_e$. The geometry of the situation is shown in Figure 2.4. The distance R represents the range of influence of the induced dipole. Outside the sphere the electrical forces may be represented by the macroscopic field and polarization, whereas inside, the detailed fields due to the other dipoles must be taken into account. Assume that the sphere is small compared with the wavelength in the medium, so that the macroscopic field is uniform across the sphere.

The field at the center of the sphere is the vector sum of three fields:

- (1) The applied field $\mathbf{E}_e$.
- (2) The field $\mathbf{E}_{ep}$ due to charges on the surface a induced by $\mathbf{E}_e$. The induced charge dq_e in an incremental surface area da is $dq_e = P_e \cos \vartheta' da$, where $da = 2\pi R^2 \sin \vartheta' d\vartheta'$, and ϑ' is the angle between $\mathbf{P}_e$ (parallel to $\mathbf{E}_e$) and the radius vector to da . The component of the field at the center parallel to $\mathbf{E}_e$, due to the charge in da is $dq_e \cos \vartheta' / 4\pi \varepsilon_{e0} R^2 = P_e \cos^2 \vartheta' da / 4\pi \varepsilon_{e0} R^2$. By symmetry, the components perpendicular to $\mathbf{E}_e$ vanish. Integrating this field over the surface of the sphere gives the contribution due to the induced surface polarization charges:

$$E_{ep} = \int_0^\pi \frac{P_e \cos^2 \vartheta' 2\pi R^2 \sin \vartheta' d\vartheta'}{4\pi \varepsilon_{e0} R^2} = \frac{P_e}{3\varepsilon_{e0}}. \quad (2.45)$$

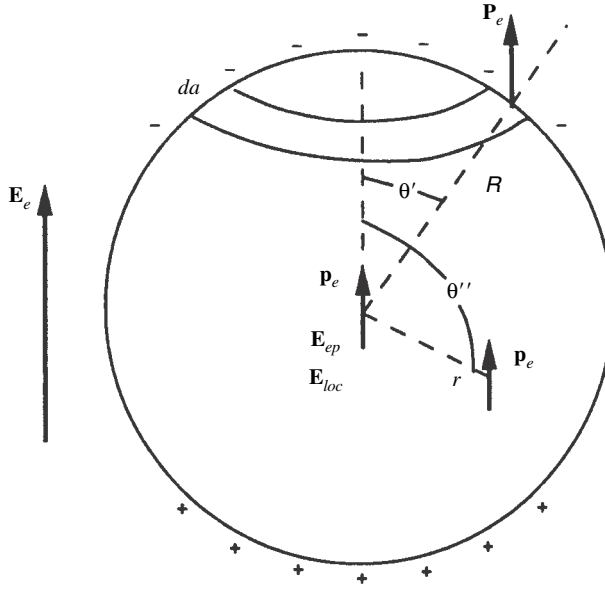


Figure 2.4

- (3) The field due to the sum of the other individual dipoles in the sphere. Assume that these are statistically distributed isotropically about the center of the cavity. Then the average component at the center due to the other dipoles may be found as follows. The potential U of a dipole of moment p_e aligned with the x -axis, at a point at distance r making an angle ϑ'' with the x -axis, is given by

$$U = p_e \cos \vartheta'' / 4\pi \epsilon_0 r^2 = p_e x / 4\pi \epsilon_0 (x^2 + y^2 + z^2)^{3/2}.$$

The x component of the field is

$$\begin{aligned} E_{ex} &= -\partial U / \partial x \\ &= -[p_e / 4\pi \epsilon_0] \left[(x^2 + y^2 + z^2)^{-3/2} - 3x^2 (x^2 + y^2 + z^2)^{-5/2} \right] \\ &= -p_e (1 - 3 \cos^2 \vartheta'') / 4\pi \epsilon_0 r^3. \end{aligned}$$

The total value of the x component of the field at the center of a sphere of radius $r < R$ due to an ensemble of such dipoles uniformly distributed over the surface of that sphere may be found by integrating the foregoing expression for E_{ex} over the surface, giving

$$\langle E_{ex} \rangle = - \int_0^\pi (p_e / 4\pi \epsilon_0 r^3) (1 - 3 \cos^2 \vartheta'') 2\pi r^2 \sin \vartheta'' d\vartheta'' = 0. \quad (2.46)$$

Hence, the field at the center due to the other dipoles in the sphere vanishes.

The total field at the center is the sum of the three contributions,

$$\mathbf{E}_{\text{loc}} = \mathbf{E}_e + \mathbf{P}_e/3\epsilon_{e0}. \quad (2.47)$$

Substituting from (2.36) and (2.37),

$$\mathbf{P}_e = N\alpha_e\epsilon_{e0}(\mathbf{E}_e + \mathbf{P}_e/3\epsilon_{e0}).$$

Solving for $\mathbf{P}_e$,

$$\mathbf{P}_e = N\alpha_e\epsilon_{e0}\mathbf{E}_e/(1 - N\alpha_e/3) = \epsilon_{e0}(K_e - 1)\mathbf{E}_e,$$

from (2.41). Solving for K_e ,

$$K_e = (1 + 2N\alpha_e/3)/(1 - N\alpha_e/3). \quad (2.48)$$

If $N\alpha_e \ll 1$, then to first order $K_e \simeq 1 + N\alpha_e$, so that $\mathbf{E}_{\text{loc}} = \mathbf{E}_e$. Equation (2.48) may be written in the alternative form

$$(K_e - 1)/(K_e + 2) = N\alpha_e/3 = N_0\rho\alpha_e/3W, \quad (2.49)$$

where N_0 is Avogadro's number ($N_0 = 6.023 \times 10^{23}$ molecules/mole), ρ is the mass density, and W is the molecular weight.

Equation (2.49) is known variously as the Clausius–Mossotti or Lorentz–Lorenz relation. Apparently it was first derived semi-empirically by R. Clausius and O. Mossotti for static fields, and later more rigorously by H. Lorentz and L. Lorenz. A similar relation was derived by Rayleigh. It describes the dependence of the dielectric constant of a gas on density reasonably well. It also gives fair agreement for many solids, as illustrated by Figure 2.5. Obviously, however, this relation is not completely general, because it predicts that K_e becomes infinite when $N\alpha_e = 3$. Further discussion of the Clausius–Mossotti relation may be found in many places, including Elliott (1966), Frohlich (1958), and Bottcher (1952).

2.3.3 Induced magnetic dipoles

In a manner analogous to the treatment of the effects of induced electric dipoles, the magnetization $\mathbf{M}_m$ is defined as the induced magnetic-dipole moment per unit volume. It will be assumed that $\mathbf{M}_m$ is linearly proportional to $\mathbf{B}_m$. The magnetic intensity $\mathbf{H}_m$ is also assumed to be proportional to $\mathbf{B}_m$ and thus to $\mathbf{M}_m$. The relationships among these quantities are

$$\mathbf{M}_m = \mu_{m0}\chi_m\mathbf{M}_m, \quad (2.50)$$

where χ_m is the magnetic susceptibility, and

$$\mathbf{B}_m = \mu_{m0}\mathbf{H}_m + \mathbf{M}_m. \quad (2.51)$$

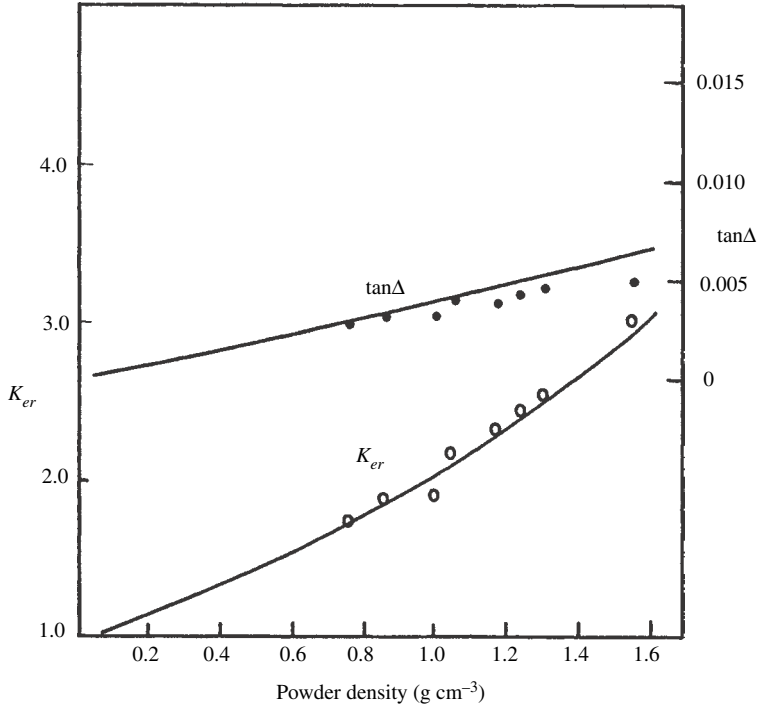


Figure 2.5 Permittivity and loss tangent as a function of density of a basalt powder finer than $37\text{ }\mu\text{m}$ at $\nu = 450\text{ MHz}$. The dots show the measured values; the lines show the Clausius–Mossotti/Lorentz–Lorenz relation normalized to the measured values of the solid material. (Reproduced from Campbell and Ulrichs [1969], copyright 1969 by the American Geophysical Union.)

Thus,

$$\mathbf{B}_m = \mu_{m0}(1 + \chi_m)\mathbf{H}_m. \quad (2.52)$$

The quantity in parentheses is the relative permeability K_m ,

$$K_m = 1 + \chi_m. \quad (2.53)$$

Then

$$\mathbf{B}_m = \mu_m \mathbf{H}_m, \quad (2.54)$$

where

$$\mu_m = \mu_{m0} K_m \quad (2.55)$$

is the *magnetic permeability*.

2.3.4 The wave equation

Equations (2.42) and (2.54), in which ε_e and μ_m are assumed to be constants for a particular material, along with $\rho_e = 0$ and $j_e = 0$, are the constitutive equations for a linear nonabsorbing medium. In such a material, Maxwell's equations, (2.1) – (2.4), reduce to the same form as (2.9) – (2.12) for free space, except that ε_{e0} and μ_{m0} are replaced by ε_e and μ_m , respectively. Reasoning identical with that in Section 2.2 then leads to equations of the form

$$\nabla^2 \mathbf{E}_e - \mu_m \varepsilon_e \partial^2 \mathbf{E}_e / \partial t^2 \quad (2.56)$$

and

$$\nabla^2 \mathbf{B}_m - \mu_m \varepsilon_e \partial^2 \mathbf{B}_m / \partial t^2. \quad (2.57)$$

These describe waves propagating with velocity

$$v = 1/\sqrt{\mu_m \varepsilon_e} = c_0/\sqrt{K_e K_m} = c_0/n \quad (2.58)$$

where n is the refractive index

$$n = \sqrt{K_e K_m}. \quad (2.59)$$

For sinusoidal waves, the frequency is determined by the source of the radiation and hence is the same in the medium and in free space. However, because $v = \lambda \nu = c/n$, the wavelength is decreased by a factor $1/n$ from its free-space value. Thus, if λ is the wavelength in free space, the field solutions are of the form

$$\mathbf{E}_e = \mathbf{E}_{e0} \exp \left[2\pi i \left(\frac{nz}{\lambda} - \nu t \right) \right] \quad (2.60)$$

and

$$\mathbf{B}_m = \mathbf{B}_{m0} \exp \left[2\pi i \left(\frac{nz}{\lambda} - \nu t \right) \right] \quad (2.61)$$

The fields $\mathbf{E}_e$ and $\mathbf{B}_m$ are related by

$$\mathbf{B}_m = \mathbf{u}_Z \times \mathbf{E}_e / v \text{ and } \mathbf{E}_e = -v \mathbf{u}_Z \times \mathbf{B}_m, \quad (2.62)$$

so that

$$B_{m0} = E_{e0}/v = nE_{e0}/c_0. \quad (2.63)$$

Then the irradiance is

$$J = \sqrt{\varepsilon_e / \mu_m} E_{e0}^2 / 2. \quad (2.64)$$

2.4 Propagation in a linear absorbing medium

2.4.1 The wave equation with finite conductivity

Absorption of radiation may be introduced into the electromagnetic equations in a number of ways. The most intuitively direct way is to assume that the medium has

a finite electrical conductivity, so that the currents induced by the incident radiation suffer losses that are described by a constitutive relation having the form of Ohm's law,

$$\mathbf{j}_e = \sigma_e \mathbf{E}_e, \quad (2.65)$$

where σ_e is the electrical conductivity.

Assume that there are no free charges, so that $\rho_e = 0$, and that the other constitutive relations are (2.42), (2.54), and (2.65). Then Maxwell's equations (2.1) – (2.4) become

$$\operatorname{div} \mathbf{E}_e = 0, \quad (2.66)$$

$$\operatorname{div} \mathbf{B}_m = 0, \quad (2.67)$$

$$\operatorname{curl} \mathbf{E}_e = -\partial \mathbf{B}_m / \partial t, \quad (2.68)$$

$$\operatorname{curl} \mathbf{B}_m = \mu_m (\sigma_e \mathbf{E}_e + \varepsilon_e \partial \mathbf{E}_e / \partial t). \quad (2.69)$$

As previously, taking the curl of (2.68), using (2.66), and substituting (2.69) gives

$$\nabla^2 \mathbf{E}_e - \mu_m \sigma_e \partial \mathbf{E}_e / \partial t - \mu_m \varepsilon_e \partial^2 \mathbf{E}_e / \partial t^2 = 0. \quad (2.70)$$

The nature of the solutions to (2.70) depends on whether the second or third term on the left-hand side dominates. The second term is of the order of $\mu_m \sigma_e \Delta E_e / \Delta t$, where ΔE_e is the change in $\mathbf{E}_e$ occurring in time Δt , while the third term is of order $\mu_m \varepsilon_e \Delta E_e / \Delta t^2$.

For periodic solutions the controlling time interval is $\Delta t \simeq 1/\nu$. Thus, if $\mu_m \sigma_e \Delta E_e \nu \ll \mu_m \varepsilon_e \Delta E_e \nu^2$, or $\sigma_e / \nu \varepsilon_e \ll 1$, the second term of (2.70) is negligible compared to the third, and the solution is essentially the wave equation. On the other hand, if $\sigma_e / \nu \varepsilon_e \gg 1$, then the second term dominates, and (2.70) is essentially the diffusion equation. Equations of the latter type describe such phenomena as the diffusion of one gas into another, the density of neutrons in a fission reactor, and heat flow. If σ_e is large, then when an external field is applied to the medium, induced charges and currents generate internal fields that are of opposite polarity to the incident field and tend to cancel it. The motion of the field appears to be retarded, and the field acts as if it were diffusing slowly into the medium rather than propagating as a wave.

2.4.2 Representations of absorption of radiation

In many types of absorptions there are no macroscopic currents, so that conductivity is an unsatisfactory way to describe the losses. Hence, absorption of electromagnetic radiation is described in the literature by a variety of parameters besides conductivity; these include imaginary component of the dielectric constant, imaginary component of the refractive index, loss tangent, optical conductivity, and the

absorption coefficient. All of these quantities are equivalent, and the choice of which particular one to use is largely a matter of taste and convention. Conductivity is usually reserved for use with metals in the frequency range where $\nu \ll \sigma_e/\epsilon_e$, complex dielectric constant and loss tangent in radio-frequency applications, and complex refractive index, optical conductivity or absorption coefficient in the ultraviolet (UV), visible, and infrared (IR) wavelength ranges.

Consider a periodically varying field of the form

$$\mathbf{E}_e(z, t) = E_e(z)e^{-2\pi i\nu t}, \quad (2.71)$$

where E_e is the amplitude of $\mathbf{E}_e$. Substituting (2.71) into (2.70) gives

$$(d^2 E_e/dz^2)e^{-2\pi i\nu t} + 2\pi i\nu t\mu_m\sigma_e E_e e^{-2\pi i\nu t} + 4\pi^2\nu^2\mu_m\epsilon_e E_e e^{-2\pi i\nu t}. \quad (2.72)$$

It is of interest to ask if a solution of the form of (2.71) may be constructed that will satisfy both equations (2.56) and (2.70). The answer is affirmative provided that the dielectric constant is allowed to be complex. That is, K_e , K_m , and σ_e are not used explicitly; instead, their effects are simply lumped together into an effective *complex dielectric constant* K_e , and the wave equation is forced to be of the form of (2.56),

$$\nabla^2 E_e - K_e\mu_{m0}\epsilon_{e0}\partial^2 E_e/\partial t^2 = 0, \quad (2.73)$$

where

$$K_e = K_{er} + iK_{ei} = K_{er}(1 + i \tan \Delta); \quad (2.74)$$

K_{er} and K_{ei} are, respectively, the real and imaginary parts of K_e , and

$$\tan \Delta = K_{ei}/K_{er} \quad (2.75)$$

is the *loss tangent*. (In many references, K_e is defined as $K_e = K_{er} - iK_{ei}$. The sign of the imaginary component makes no practical difference as long as the same convention is maintained throughout.)

Putting (2.71) into (2.72) gives

$$(d^2 E_e/dz^2)e^{-2\pi i\nu t} + 4\pi^2\nu^2 K_e\mu_{m0}\epsilon_{e0} E_e e^{-2\pi i\nu t}. \quad (2.76)$$

Comparing this last equation with (2.72) shows that in order for the two equations to be equivalent, the following must hold:

$$K_{er} = K_e K_m, \quad (2.77)$$

$$K_{ei} = K_m\sigma_e/2\pi\nu\epsilon_{e0}. \quad (2.78)$$

Solving for σ_e gives

$$\sigma_e = 2\pi\nu\epsilon_{e0}K_{ei}/K_m. \quad (2.79)$$

Equation (2.79) can be used to calculate an effective σ_e from measured values of K_{ei} and K_m . The conductivity defined in this way is called the *optical conductivity*. Except for the special case of ferromagnetic materials, μ_m is usually not significantly different from μ_0 , and $K_m = 1$.

If the dielectric constant is complex, then the refractive index is also. From (2.45),

$$n = \sqrt{K_e} = n_r + in_i, \quad (2.80)$$

where n_r and n_i are the real and imaginary parts, respectively, of the *complex refractive index* n . (If K_e is defined to be $K_e = K_{er} - iK_{ei}$, then m must be defined as $m = n - ik$.)

Combining (2.74) and (2.80),

$$K_e = K_{er} + iK_{ei} = n^2 = (n_r + in_i)^2 = n_r^2 - n_i^2 + 2in_r n_i. \quad (2.81)$$

Thus,

$$K_{er} = n_r^2 - n_i^2, \quad (2.82)$$

and

$$K_{ei} = 2n_r n_i. \quad (2.83)$$

Solving for n_r and n_i ,

$$n_r = \left\{ \frac{1}{2} \left[(K_{er}^2 + K_{ei}^2)^{1/2} + K_{er} \right] \right\}^{1/2}, \quad (2.84)$$

and

$$n_i = \left\{ \frac{1}{2} \left[(K_{er}^2 + K_{ei}^2)^{1/2} - K_{er} \right] \right\}^{1/2}. \quad (2.85)$$

If $\tan \Delta \ll 1$, these become

$$n_r \approx \sqrt{K_{er}}, \quad (2.86)$$

$$n_i \approx K_{ei} / 2\sqrt{K_{er}}. \quad (2.87)$$

At this point the reader may well be wondering how a refractive index can be complex, because this also would seem to imply a complex wavelength and propagation velocity. One answer is that this is simply a convenient mathematical representation; the imaginary part can always be recast into a form that describes a real quantity. To see the physical meaning and implications of a complex dielectric constant or refractive index, write the solution to the wave equation explicitly in the form of (2.60), which describes a plane wave propagating in the positive z direction in a medium of complex refractive index n :

$$\mathbf{E}_e = \mathbf{E}_{e0} e^{2\pi i(nz/\lambda - vt)} = \mathbf{E}_{e0} e^{2\pi i[(n_r + in_i)z/\lambda - vt]} = \mathbf{E}_{e0} e^{-2\pi n_i z/\lambda} e^{2\pi i(n_r z/\lambda - vt)}. \quad (2.88)$$

Equation (2.88) represents a sinusoidal wave with frequency ν , wavelength λ/n_r and velocity $v = c_0/n_r$, where λ is the wavelength in free space, propagating through the medium with exponentially decreasing amplitude. The magnetic field has the same form,

$$\mathbf{B}_m = \mathbf{B}_{m0} e^{-2\pi n_i z/\lambda} e^{2\pi i (n_r z/\lambda - \nu t)}, \quad (2.89)$$

where, from (2.63),

$$\mathbf{B}_{m0} = (n_r + in_i) E_{e0}/c_0. \quad (2.90)$$

Now, the actual electric field is

$$\text{Re}(\mathbf{E}_e) = E_{e0} e^{-2\pi n_i z/\lambda} \cos[2\pi (n_r z/\lambda - \nu t)], \quad (2.91)$$

and the actual magnetic field is

$$\begin{aligned} \text{Re}(\mathbf{B}_m) &= \text{Re} \left[(n_r + in_i) (E_{e0}/c_0) e^{-2\pi n_i z/\lambda} e^{2\pi i (n_r z/\lambda - \nu t)} \right] \\ &= (E_{e0}/c_0) e^{-2\pi n_i z/\lambda} \{ n_r \cos[2\pi (n_r z/\lambda - \nu t)] \\ &\quad - n_i \sin[2\pi (n_r z/\lambda - \nu t)] \} \end{aligned} \quad (2.92)$$

This can also be written

$$\text{Re}(\mathbf{B}_m) = (E_{e0}/c_0) e^{2\pi n_i z/\lambda} n_r \cos[2\pi (n_r z/\lambda - \nu t + \psi)] \quad (2.93)$$

where $\tan \psi = n_i/n_r$. Comparing (2.91) and (2.93) shows that in an absorbing medium the phases of the electric and magnetic fields are no longer the same, but differ by angle ψ .

The irradiance is given by $J = \langle \text{Re}(\mathbf{E}_e) \times \text{Re}(\mathbf{B}_m) / \mu_{m0} \rangle$, where μ_m has been set equal to μ_{m0} in the spirit of lumping all the propagation constants into K_e , and the angular brackets denote a time average over one period. Then the integration over time is readily carried out to give

$$J = \text{Re} \left(n |\mathbf{E}_e|^2 / 2\mu_{m0} \right) = \frac{1}{2} \sqrt{\epsilon_{e0}/\mu_{m0}} n_r E_{e0}^2 e^{-4\pi n_i z/\lambda}. \quad (2.94)$$

Note that only the portion of $\mathbf{B}_m$ that is in phase with $\mathbf{E}_e$ contributes to the irradiance.

Expression (2.94) for J is of the form $J(z) = J(0) e^{-4\pi n_i z/\lambda}$. This has the same form as Beer's law,

$$J(z) = J(0) e^{-\alpha z}, \quad (2.95a)$$

where α is the *absorption coefficient*,

$$\alpha = 4\pi n_i / \lambda. \quad (2.95b)$$

Equation (2.95b) is often called the *dispersion relation*, although, as will be seen in Chapter 3, it is but one of several dispersion relations. It shows that the irradiance is reduced by a factor of e when the wave has propagated a distance $l_e = \alpha^{-1} = \lambda/4\pi n_i$. This distance is called the *absorption length*.

2.5 Interference

Suppose that two electromagnetic waves $E_1 = E_{10} \cos 2\pi(z/\lambda_1 - v_1 t + \phi_1)$ and $E_2 = E_{20} \cos 2\pi(z/\lambda_2 - v_2 t + \phi_2)$ are propagating in the same direction, where ϕ_1 and ϕ_2 are the phases of the two waves at point z and time t , and that the fields are parallel to each other. We have seen that irradiance is proportional to the time average of the square of the electric field, $\langle (E_1 + E_2)^2 \rangle$, which is

$$\begin{aligned}
 & \frac{1}{T} \int_0^T E_{10}^2 \cos^2 2\pi \left(\frac{z}{\lambda_2} - v_2 t + \phi_2 \right) dt + \frac{1}{T} \int_0^T E_{20}^2 \cos^2 2\pi \left(\frac{z}{\lambda_2} - v_2 t + \phi_2 \right) dt \\
 & + \frac{1}{T} \int_0^T 2E_{10}E_{20} \cos 2\pi \left(\frac{z}{\lambda_1} - v_1 t + \phi_1 \right) \cos 2\pi \left(\frac{z}{\lambda_2} - v_2 t + \phi_2 \right) dt \\
 & = \frac{E_{10}^2}{2T} \int_0^T \left[1 + \cos 4\pi \left(\frac{z}{\lambda_1} - v_1 t + \phi_1 \right) \right] dt + \frac{E_{20}^2}{2T} \int_0^T \\
 & \times \left[1 + \cos 4\pi \left(\frac{z}{\lambda_2} - v_2 t + \phi_2 \right) \right] dt \\
 & + \frac{E_{10}E_{20}}{T} \int_0^T \left[\cos 2\pi \left(\frac{z}{\lambda_1} - v_1 t + \phi_1 + \frac{z}{\lambda_2} - v_2 t + \phi_2 \right) \right. \\
 & \left. + \cos 2\pi \left(\frac{z}{\lambda_1} - v_1 t + \phi_1 - \frac{z}{\lambda_2} + v_2 t - \phi_2 \right) \right] dt \quad (2.96)
 \end{aligned}$$

Now, all of the cosine functions in the integrands of (2.69) are oscillating functions of t with average values of zero. If $v_1 \neq v_2$ their integrals become vanishingly small over times T long compared with $1/v_1$ and $1/v_2$, so that the total integral is proportional to $E_{10}^2 + E_{20}^2$. Thus the total irradiance is equal to the sum of the individual irradiances. However, if $v_1 = v_2$ ($\lambda_1 = \lambda_2$), then the last integral becomes

$$\frac{E_{10}E_{20}}{T} \int_0^T \left[\cos 2\pi \left(\frac{2z}{\lambda_1} - 2v_1 t + \phi_1 + \phi_2 \right) + \cos 2\pi(\phi_1 - \phi_2) \right] dt. \quad (2.97)$$

The integral over the first, but not the second, cosine vanishes, so that now the integral over time is proportional to $E_{10}^2 + 2E_{10}E_{20} \cos 2\pi(\phi_1 - \phi_2) + E_{20}^2$. Hence the total irradiance now depends on the difference in phase between the two waves. If the two waves are in phase, $\phi_1 = \phi_2$, the integral over time is proportional to $(E_{10} + E_{20})^2$, and the total irradiance is proportional to the square of the sum of the amplitudes of the individual waves. The waves are said to *interfere positively*. However, if $2\pi(\phi_1 - \phi_2) = \pi$, the integral over time is proportional to $(E_{10} - E_{20})^2$, the total irradiance is proportional to the square of the difference between the amplitudes of the individual waves, and the waves are said to *interfere negatively*. In that case if $E_{10} = E_{20}$ the two waves exactly cancel each other out and the irradiance is zero.

2.6 Polarization; the Stokes vector

2.6.1 Linear and circular polarization

An electromagnetic wave propagating in the z direction can be resolved into two waves with electric vectors parallel to the x and y axes: $\mathbf{E}_x = \mathbf{u}_x E_{x0} \cos(2\pi x/\lambda - 2\pi \nu t + \phi_x)$ and $\mathbf{E}_y = \mathbf{u}_y E_{y0} \cos(2\pi x/\lambda - 2\pi \nu t + \phi_y)$, where $\mathbf{u}_x$ and $\mathbf{u}_y$ are unit vectors, and ϕ_x and ϕ_y are the phases of the fields at point $x = 0$ and time $t = 0$. These axes are perpendicular to each other, but their directions are otherwise arbitrary and can be chosen to fit the situation of interest. Suppose $\phi_x = \phi_y = 0$. Then at any point z , say $z = 0$, the two components will oscillate in phase; that is, both will increase, go through maximum, decrease, reverse, and so on, at the same time. The resultant combination of the two vectors will be an electric field that always oscillates in the same plane making an angle with the x and y axes that is determined by the ratio of E_{y0} to E_{x0} . The light is said to be *linearly polarized*.

However, suppose the two waves are not in phase. Suppose $\phi_y = 0$, and $\phi_x = -\pi/2$; that is, the x field lags behind the y field by 90° . Then at $t = 0$ the y field points in the positive y direction and is at its maximum, but the x field is zero. As time proceeds the y field decreases while the x field grows in the positive x direction until one quarter of a period later the y field is zero while the x field is maximum. If the resultant of the combined fields is represented by a vector, the vector has rotated in the clockwise direction and the tip has traced out one quarter of an ellipse. During one period the tip traces out a complete ellipse, so the light is said to be *elliptically polarized*.

If $E_{y0} = E_{x0}$ the tip of the electric vector traces out a circle and the light is said to be *circularly polarized*. A little thought will show that an elliptically polarized wave can be represented by the sum of two circularly polarized waves rotating in opposite directions. If $\phi_y - \phi_x < 0$ the electric vector rotates in the clockwise (CW) direction and the light is said to be *right-handed circularly polarized* or have *right-handed helicity*. If $\phi_y - \phi_x > 0$ the vector rotates in the counterclockwise (CCW) direction and the light is said to be *left-handed circularly polarized* or have *left-handed helicity*.

Unfortunately there is an ambiguity over the definition of *handedness* or *helicity*. Historically, workers in the optical region of the spectrum have defined the helicity as that when the observer is looking in the direction from which the radiation is coming, but persons working at radio frequencies have defined the helicity as that when looking in the direction into which the radiation is propagating. In this book we will follow the latter convention and define CW (right-handed) and CCW (left-handed) rotation as that seen by an observer looking in the direction into which the radiation is propagating. Thus, a circularly polarized wave has right-handed helicity

if a screw with a right-handed thread would advance in the same direction as the direction of propagation of the wave if turned CW. When consulting a reference the convention used should always be ascertained.

2.6.2 The Jones and Stokes vectors

A propagating electromagnetic wave may be conveniently described in matrix form by the *Jones vector*. Suppose the wave is propagating along the z -axis in the positive z direction. Then in matrix notation the general form of the Jones vector for the electric field is

$$\mathbf{E} = \begin{pmatrix} E_{x0} \exp[2\pi i(z/\lambda - \nu t) + i\phi_x] \\ E_{y0} \exp[2\pi i(z/\lambda - \nu t) + i\phi_y] \end{pmatrix}. \quad (2.98a)$$

Then the intensity of the wave is proportional to $|\mathbf{E}|^2 = E_{x0}^2 + E_{y0}^2$. Since the absolute phase is usually not of interest, the Jones vector is often simplified by dividing by the common factor $\exp[2\pi i(z/\lambda - \nu t)]$, so that the simplified Jones vector contains only amplitude and phase difference:

$$\mathbf{E} = \begin{pmatrix} E_{x0} \exp(i\phi_x) \\ E_{y0} \exp(i\phi_y) \end{pmatrix}. \quad (2.98b)$$

Finally it is common practice to normalize the vector by dividing by $\sqrt{E_{x0}^2 + E_{y0}^2} \exp(i\phi_x)$, so that the *normalized Jones vector* is

$$\mathbf{E} = \begin{pmatrix} \frac{E_{x0}}{\sqrt{E_{x0}^2 + E_{y0}^2}} \\ \frac{E_{y0}}{\sqrt{E_{x0}^2 + E_{y0}^2}} e^{i(\phi_y - \phi_x)} \end{pmatrix}. \quad (2.98c)$$

Examples of the normalized Jones vector are given in Table 2.1.

Light may be *unpolarized*, in which case its electric vectors point randomly in all directions perpendicular to the direction of propagation, or it may be a combination of unpolarized, linearly polarized, or circularly polarized radiation. The difficulty with the Jones vector is that it cannot describe unpolarized light. If the light is partially polarized it may be described by a quantity known as the *Stokes vector*, which has four components and is usually denoted by

$$S = \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}. \quad (2.99)$$

Table 2.1. *Some examples of the normalized Jones and Stokes vectors*
Polarization state Jones vector Stokes vector

Polarization state	Jones vector	Stokes vector
Unpolarized		$I \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$
Linearly polarized along x -axis	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$I \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$
Linearly polarized along y -axis	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$I \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$
Linearly polarized 45° CCW from x -axis	$\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$	$I \begin{pmatrix} 1 \\ 0 \\ +1 \\ 0 \end{pmatrix}$
Linearly polarized 45° CW from x -axis	$\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$	$I \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$
Left-handed circularly polarized	$\begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}$	$I \begin{pmatrix} 1 \\ 0 \\ 0 \\ +1 \end{pmatrix}$
Right-handed circularly polarized	$\begin{pmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}$	$I \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$

Denoting the time average of a quantity by angular brackets, the components of the Stokes vector are defined as follows:

$$I = C(< \mathbf{E}_x^2 > + < \mathbf{E}_y^2 >), \quad (2.100a)$$

$$Q = C(< \mathbf{E}_x^2 > - < \mathbf{E}_y^2 >), \quad (2.100b)$$

$$U = C(< 2\mathbf{E}_x\mathbf{E}_y \cos(\varnothing_y - \varnothing_x) >), \quad (2.100c)$$

$$V = C(< 2\mathbf{E}_x\mathbf{E}_y \sin(\varnothing_y - \varnothing_x) >), \quad (2.100d)$$

where $C = \frac{1}{2}\sqrt{\epsilon_{e0}/\mu_{m0}n_r}$. Physically, I is the total intensity; Q is the linearly polarized intensity parallel to the x -axis minus that parallel to the y -axis; U is the linearly polarized intensity along a direction rotated 45° CCW from the x -axis minus that along a direction rotated 45° CW from the x -axis; V is the left-handed polarized intensity minus the right-handed polarized intensity. Often the Stokes vector is normalized by dividing all quantities by I , in which case it is written

$$S = I \begin{pmatrix} 1 \\ Q \\ U \\ V \end{pmatrix}, \quad (2.101)$$

where Q , U , and V have values between -1 and $+1$. Since I is the total intensity the components must satisfy the relation: $Q + U + V \leq 1$. Examples of the normalized Stokes vector are given in Table 2.1.

2.6.3 Polarization

The *degree of polarization*, also often called the *linear polarization ratio*, and just plain *polarization*, is defined as

$$P = -Q/I, \quad (2.102)$$

and is the difference between the linearly polarized intensity along two perpendicular axes divided by the total intensity.